

SANDESH NARENDRA BAGMARE

Wadmukhwadi, Pune

+91-8975963453 sandeshbagmare2110@gmail.com [LinkedIn](#)



EDUCATION

Institution	Degree/Exam	Year	Score
Vishwakarma Institute of Technology, Pune	B.Tech E&TC	2021 – 2025	8.84 CGPA
Army Public School Kirkee, Pune	CBSE HSC	2021	90.80%
Army Public School Kirkee, Pune	CBSE SSC	2019	91.83%

EXPERIENCE

PTC

Associate Software Analyst

July 2025 – Present

Pune, India

- Developed **Twx AI Assistant Agentic Framework** with multi-agent PoC using LangGraph and Plan-and-Act methodology
- Built **AI Agents Evaluation Framework** integrating Azure AI Evaluation, DeepEval, and 4 custom evaluators with HTML/Allure reporting
- Engineered **CLI-based Agent Development Kit (ADK)** for automated agent creation with customization capabilities
- Enhanced **Twx Copilot RAG system** with Qdrant-based pipelines, optimized chunking, and hybrid reranking for improved retrieval
- Contributed to development of agents and A2A capabilities for framework

PTC

Software Engineering Intern

July 2024 – June 2025

Pune, India

- Developed comprehensive **unit tests for Helm charts** across Solution Central repositories ensuring deployment reliability
- Resolved **security vulnerabilities** identified by Invicti, Prisma, and Black Duck for Solution Central 3.1.2 release
- Created **Terraform scripts** for Azure Flexible Server 16 deployment and PostgreSQL 11 migration validation scripts
- Designed **LangGraph flow architecture** for Twx Copilot ensuring data privacy and optimized token usage
- Developed and presented **Twx Copilot RAG-based PoC** demonstrating intelligent product assistance capabilities

PROJECTS

- AI-Powered Social Media Automation Platform | Generative AI, Python, Analytics2025
- Automated AI-driven content creation, scheduling, and publishing across Instagram and YouTube accounts
 - Optimized SEO using LLM-generated hashtags, captions, and performance-based content pruning
 - Built analytics pipeline for engagement tracking, reach prediction, and quality thresholding
 - Scaled platform to manage 10+ accounts via API-based automation and influencer outreach
- Vision-Based Humanoid Robotic Palm | Deep Learning, ResNet, YOLOv5, VGG16, Arduino2024
- Developed vision-based robotic palm to assist amputees in performing daily tasks with Arduino Uno for precise gripping control
 - Achieved high object detection accuracies: ResNet (98.92%), VGG16 (96.55%), YOLOv5 (84.90%)
 - Integrated deep learning classification with hardware control for real-time object recognition and adaptive gripping
- Pressure Sensor based Smart Bed | Python, FSR Sensors, MySQL, Data Visualization2024
- Developed smart bed with 20x10 FSR sensor matrix to monitor body pressure distribution during sleep with real-time color-coded maps
 - Integrated MySQL database for pressure data storage with timestamps enabling future analysis and research applications
 - Designed for bed sore prevention, sleep tracking, fall detection (elderly/infants), and sleep disorder diagnosis

TECHNICAL SKILLS

- Languages:** Java, Python, HTML, CSS, JavaScript, Bash
- Cloud & DevOps:** Helm, Terraform, Azure, Git, Jenkins
- AI/ML Technologies:** LangChain, LangGraph, RAG, Qdrant, Azure AI Evaluation, DeepEval
- Security Tools:** BurpSuite, Invicti, Prisma, Black Duck
- Databases:** PostgreSQL, Azure Flexible Server
- Core Concepts:** Agentic AI, Multi-Agent Architectures, DSA, OOP, DBMS, OS, Cybersecurity

AWARDS AND RECOGNITIONS

- Product Spotlight Winner Q3'2025: ThingWorx AI contributions
- Team PTC: AI contributions
- TCS Cybersecurity HackQuest9 CTF: Digital Package secured
- Team PTC: Terraform and Security issues for SC 3.1.2
- Brain, Passion, and Fun: Helm contribution

CERTIFICATIONS

- AI & Cybersecurity 2025: O'Reilly
- LinkedIn AI Academy: LinkedIn Learning
- Google Cybersecurity: Coursera
- Securing Generative AI: O'Reilly
- Agentic AI in Action: O'Reilly